Présentation CCP Développeur Web & Web mobile Partie Back End

20/11/2025

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Contexte du projet : Introduction

PokeRNCP est une API REST de type Pokédex, elle permet aux utilisateurs de créer un compte et de collectionner les Pokémon des 6 premières générations.

Contexte du projet : Equipe

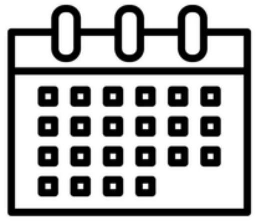
Xabi Martinez

32 ans

Holberton C#24

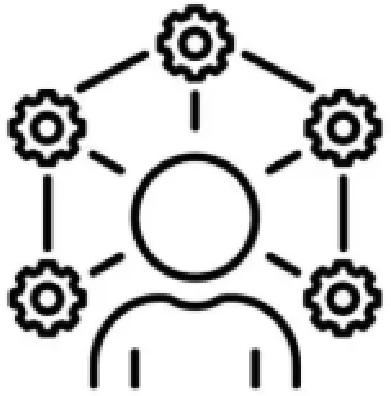
Bordeaux

Contexte du projet : Contraintes



Temps

Organisation



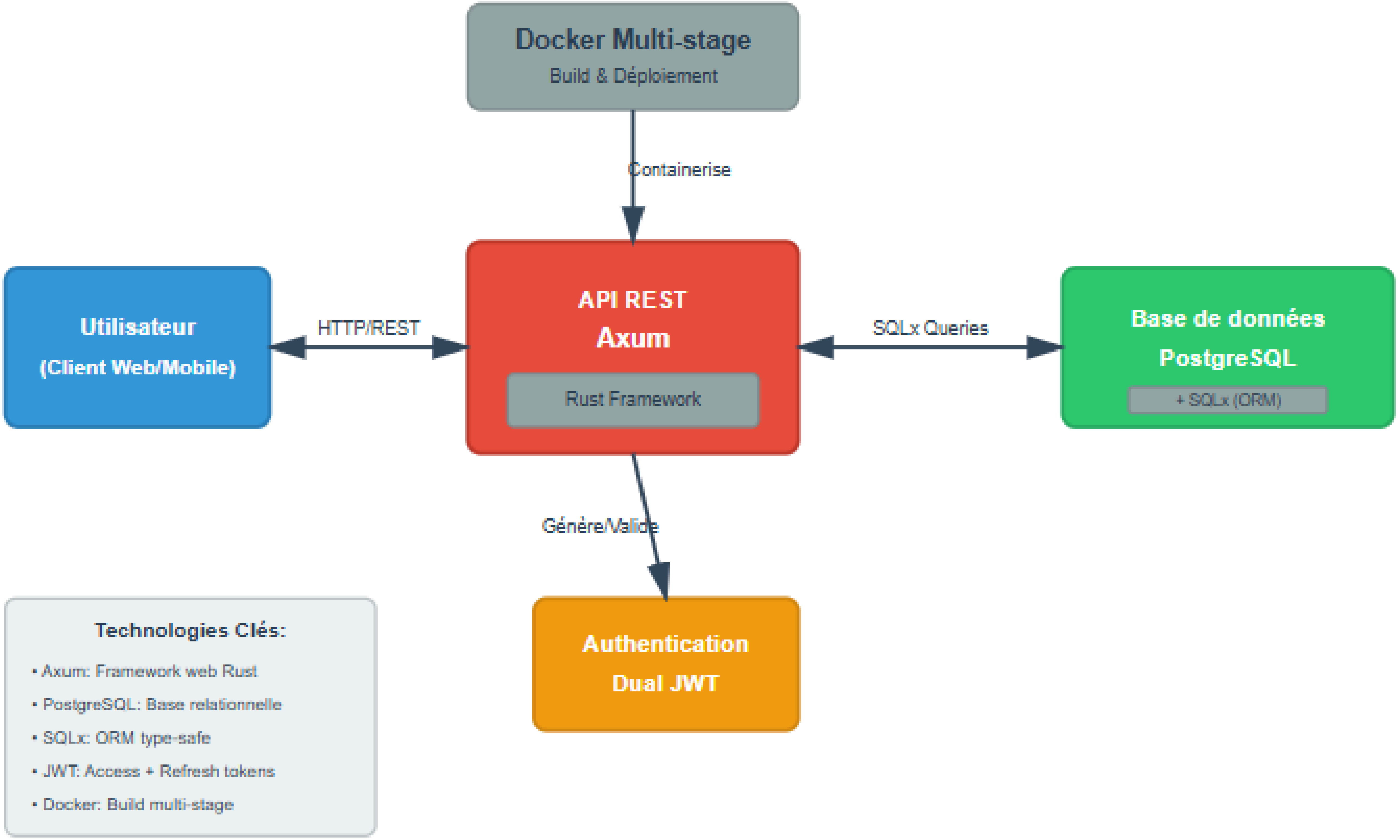
Compétences

Contexte du projet : Utilisation

- CRUD de compte
- Capture de Pokémon
- Data sur un Pokémon capturé
- Possibilité de trouver une Pokémon spécifique

Gestion du projet : Outils

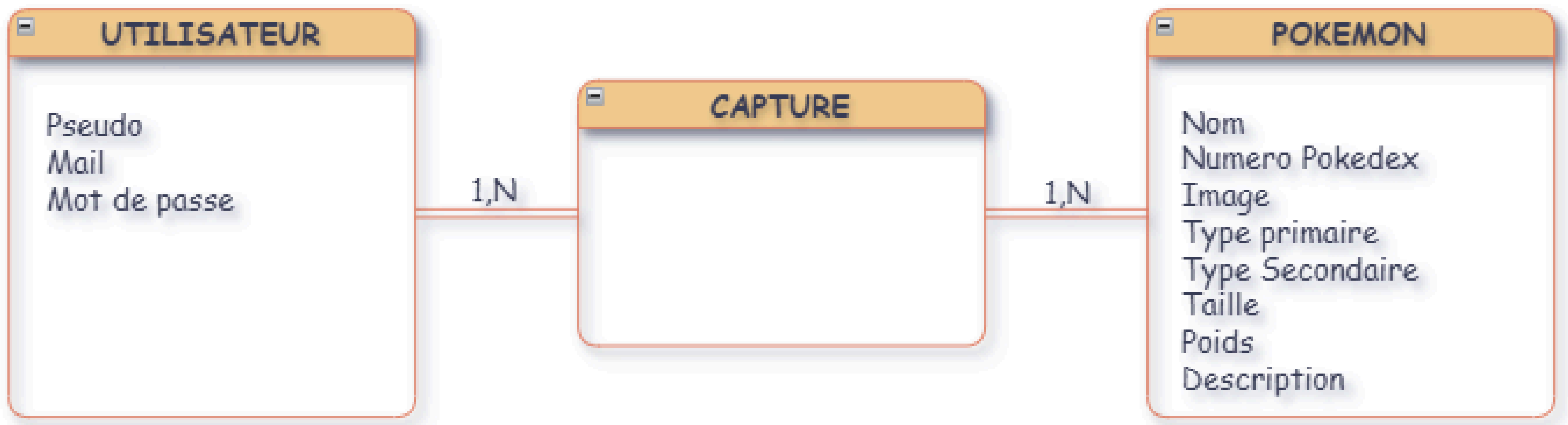
Architecture globale



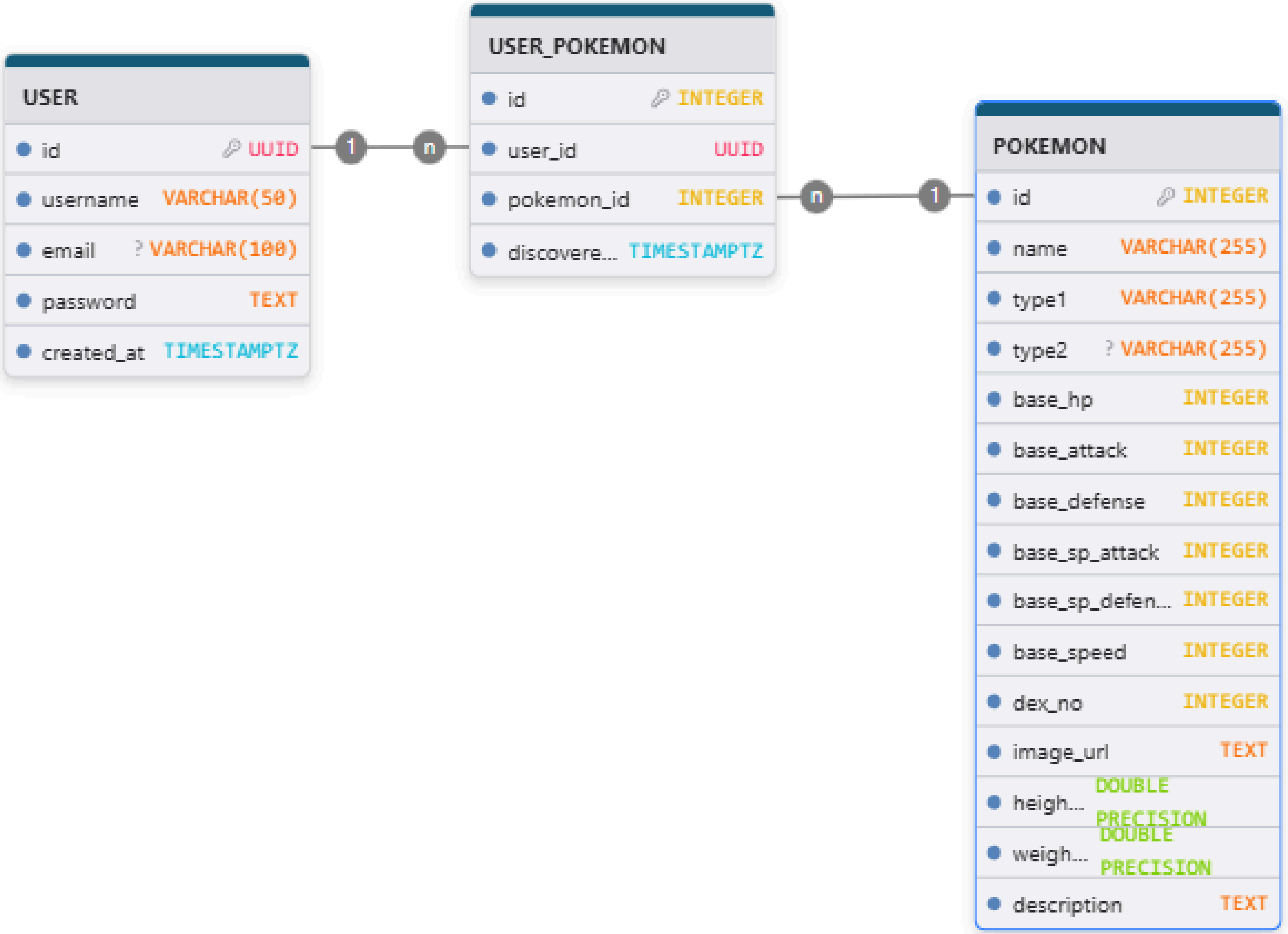
Methode Merise

- MCD : Users, Pokémon, User_Pokemon
- Relation N<=>N capturée
- Attribut dérivé caught
- Contraintes UNIQUE + cascade
- Modèle stable & normalisé

MCD


UTILISATEUR
Pseudo
Mail
Mot de passe
1,N
CAPTURE
1,N
POKEMON
Nom
Numero Pokedex
Image
Type primaire
Type Secondaire
Taille
Poids
Description

MPD


USER
id UUID
username VARCHAR(50)
email ? VARCHAR(100)
password TEXT
created_at TIMESTAMPTZ
1
n
USER_POKEMON
id INTEGER
user_id UUID
pokemon_id INTEGER
discovere... TIMESTAMPTZ
n
1
POKEMON
id INTEGER
name VARCHAR(255)
type1 VARCHAR(255)
type2 ? VARCHAR(255)
base_hp INTEGER
base_attack INTEGER
base_defense INTEGER
base_sp_attack INTEGER
base_sp_defen... INTEGER
base_speed INTEGER
dex_no INTEGER
image_url TEXT
heigh... DOUBLE PRECISION
weigh... DOUBLE PRECISION
description TEXT

Initialisation DB

init_db

```sql
-- Requis pour gen_random_uuid()
CREATE EXTENSION IF NOT EXISTS pgcrypto;

-- ===== TABLE USERS =====
CREATE TABLE IF NOT EXISTS users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(100) UNIQUE,
    password TEXT NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- ===== TABLE POKEMON =====
CREATE TABLE IF NOT EXISTS pokemon (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) UNIQUE NOT NULL,
    type1 VARCHAR(20) NOT NULL,
    type2 VARCHAR(20),
    base_hp INTEGER,
    base_attack INTEGER,
    base_defense INTEGER,
    base_sp_attack INTEGER,
    base_sp_defense INTEGER,
    base_speed INTEGER
);

-- ===== TABLE ASSOCIATIVE USER_POKEMON =====
CREATE TABLE IF NOT EXISTS user_pokemon (
    id SERIAL PRIMARY KEY,
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    pokemon_id INTEGER NOT NULL REFERENCES pokemon(id),
    nickname VARCHAR(50),
    discovered_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    UNIQUE(user_id, pokemon_id)
);

-- ===== INDEX =====
CREATE INDEX IF NOT EXISTS idx_user_pokemon_user_id ON user_pokemon(user_id);
CREATE INDEX IF NOT EXISTS idx_user_pokemon_pokemon_id ON user_pokemon(pokemon_id);
```

Migration DB

migration

```sql
-- Migration: add_pokemon_image_and_number (UP)
-- Ajoute des colonnes pour stocker le numéro du Pokédex et l'URL de l'image

ALTER TABLE pokemon
    ADD COLUMN IF NOT EXISTS dex_no INTEGER,
    ADD COLUMN IF NOT EXISTS image_url TEXT;
```

migration

```sql
-- Migration: add_pokemon_measurements_and_weaknesses (UP)
-- Ajoute taille/poids (en unités SI) et faiblesses

ALTER TABLE pokemon
    ADD COLUMN IF NOT EXISTS height_m DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS weight_kg DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS weaknesses TEXT[];
```

migration

```sql
-- Drop weaknesses column and add description to pokemon
ALTER TABLE pokemon
    DROP COLUMN IF EXISTS weaknesses,
    ADD COLUMN IF NOT EXISTS description TEXT;
```

Sécurité de l’API

- JWT access court / refresh long
- Cookies HttpOnly, SameSite Lax/Strict
- Argon2 pour les mots de passe
- Middlewares Axum
- Mitigation CSRF (cookies + header)

Flux applicatifs

- Login : validation + tokens
- Refresh automatique
- Listing Pokémon
- Capture (idempotente)
- Vérifications & permissions

Extraits de code

catch function

```rust
pub async fn catch(
    CurrentUser(user_id): CurrentUser,
    State(pool): State<PgPool>,
    Json(payload): Json<CatchByNamePayload>,
) -> ApiResult<(axum::http::StatusCode, String)> {
    let pokemon_id = sqlx::query_scalar::<_, i32>(r#"SELECT id FROM pokemon WHERE name = $1"#)
        .bind(&payload.name)
        .fetch_optional(&pool)
        .await
        .map_err(to_500)?
        .ok_or_else(|| not_found("Pokémon introuvable."))?;

    let _ = sqlx::query(
        r#"
        INSERT INTO user_pokemon (user_id, pokemon_id, nickname)
        VALUES ($1, $2, $3)
        ON CONFLICT (user_id, pokemon_id) DO NOTHING
        "#,
    )
    .bind(user_id)
    .bind(pokemon_id)
    .bind(payload.nickname)
    .execute(&pool)
    .await
    .map_err(to_500)?;

    created("Pokémon marqué comme capturé.")
}
```

list_all function

```rust
pub async fn list_all(
    CurrentUser(user_id): CurrentUser,
    State(pool): State<PgPool>,
) -> ApiResult<Json<Vec<PokemonWithCaught>>> {
    let rows = sqlx::query_as::<_, PokemonWithCaught>(
        r#"
        SELECT
            p.id          AS id,
            p.name        AS name,
            p.type1       AS type1,
            p.type2       AS type2,
            p.dex_no      AS dex_no,
            p.image_url   AS image_url,
            EXISTS (
                SELECT 1 FROM user_pokemon up
                WHERE up.user_id = $1 AND up.pokemon_id = p.id
            )             AS caught
        FROM pokemon p
        ORDER BY p.id
        "#,
    )
    .bind(user_id)
    .fetch_all(&pool)
    .await
    .map_err(to_500)?;

    Ok(Json(rows))
}
```

Performance et optimisations

- Rust + Tokio
- Requêtes compilées SQLx
- Démarrage : retry DB
- Idempotence et LIMIT
- Calcul dynamique caught

Qualité et tests

- Tests d’intégration (reqwest)
- nextest : exécution rapide
- Linter clippy
- Séparation clean du code
- Documentation claire

Tests

db_test function

```rust
#[tokio::test]
async fn la_base_de_donnees_repond() {
    let pool = common::test_pool().await;
    let one: i32 = sqlx::query_scalar("SELECT 1")
        .fetch_one(pool)
        .await
        .unwrap();
    assert_eq!(one, 1);
}
```

unique_user_test function

```rust
#[tokio::test]
async fn create_user_201_et_conflit_409() {
    let (base, handle) = start_server().await;
    let client = reqwest::Client::new();
    let unique = uuid::Uuid::new_v4().to_string();
    let username = format!("user_{}", unique);
    let email = format!("{}@example.com", username);
    let res = client
        .post(format!("{}/api/users", base))
        .json(&json!({
            "username": username,
            "email": email,
            "password": "Password123!"
        }))
        .send()
        .await
        .unwrap();
    assert_eq!(res.status(), StatusCode::CREATED);

    let res = client
        .post(format!("{}/api/users", base))
        .json(&json!({
            "username": username,
            "email": email,
            "password": "Password123!"
        }))
        .send()
        .await
        .unwrap();
    assert_eq!(res.status(), StatusCode::CONFLICT);

    handle.abort();
    delete_user(&username).await;
}
```

Tests

```
Nextest run ID 30847ad9-3b4b-483a-a030-82761db22e59 with nextest profile: default
    Starting 17 tests across 8 binaries
        PASS [   0.709s] pokedex_rncp_backend::health_test get_root_retourne_bienvenue
        PASS [   0.710s] pokedex_rncp_backend::db_test la_base_de_donnees_repond
        PASS [   0.718s] pokedex_rncp_backend::auth logout_definit_cookies_expirants
        PASS [   0.712s] pokedex_rncp_backend::health_test path_inconnu_retourne_404
        PASS [   0.727s] pokedex_rncp_backend::pokemon list_requiert_auth
        PASS [   0.744s] pokedex_rncp_backend::auth me_requiert_authentification
        PASS [   0.741s] pokedex_rncp_backend::auth tokens_invalides_donnent_401
        PASS [   1.503s] pokedex_rncp_backend::auth me_ok_avec_bearer
        PASS [   1.542s] pokedex_rncp_backend::auth me_ok_avec_cookie_auth
        PASS [   1.554s] pokedex_rncp_backend::auth refresh_token_ok_avec_cookie_refresh
        PASS [   1.555s] pokedex_rncp_backend::auth refresh_token_requiert_bearer_et_definit_cookie
        PASS [   1.130s] pokedex_rncp_backend::pokemon list_search_catch_get
        PASS [   1.172s] pokedex_rncp_backend::user delete_user_soi_meme
        PASS [   1.604s] pokedex_rncp_backend::user create_user_201_et_conflit_409
        PASS [   1.890s] pokedex_rncp_backend::user update_user_refuse_si_different
        PASS [   2.219s] pokedex_rncp_backend::user update_user_soi_meme_modifie_username_email_et_password
        PASS [   3.182s] pokedex_rncp_backend::auth reset_mot_de_passe_retourne_token_et_confirmation_mise_a_jour
```

Crates utilisés

Tokio

A runtime for writing reliable, asynchronous, and slim applications with the Rust programming language. It is:

- **Fast**: Tokio's zero-cost abstractions give you bare-metal performance.
- **Reliable**: Tokio leverages Rust's ownership, type system, and concurrency model to reduce bugs and ensure thread safety.
- **Scalable**: Tokio has a minimal footprint, and handles backpressure and cancellation naturally.

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Website | Guides | API Docs | Chat

Overview

Tokio is an event-driven, non-blocking I/O platform for writing asynchronous applications with the Rust programming language. At a high level, it provides a few major components:

- A multithreaded, work-stealing based task scheduler.
- A reactor backed by the operating system's event queue (epoll, kqueue, IOCP, etc.).
- Asynchronous TCP and UDP sockets.

These components provide the runtime components necessary for building an asynchronous application.

RustCrypto: Argon2

crate | docs passing | argon2 no status | license Apache2.0/MIT | rustc 1.65+ | zulip join chat

Pure Rust implementation of the Argon2 password hashing function.

Documentation

About

Argon2 is a memory-hard key derivation function chosen as the winner of the Password Hashing Competition in July 2015.

It implements the following three algorithmic variants:

- **Argon2d**: maximizes resistance to GPU cracking attacks
- **Argon2i**: optimized to resist side-channel attacks
- **Argon2id**: (default) hybrid version combining both Argon2i and Argon2d

Support is provided for embedded (i.e. `no_std`) environments, including ones without `alloc` support.

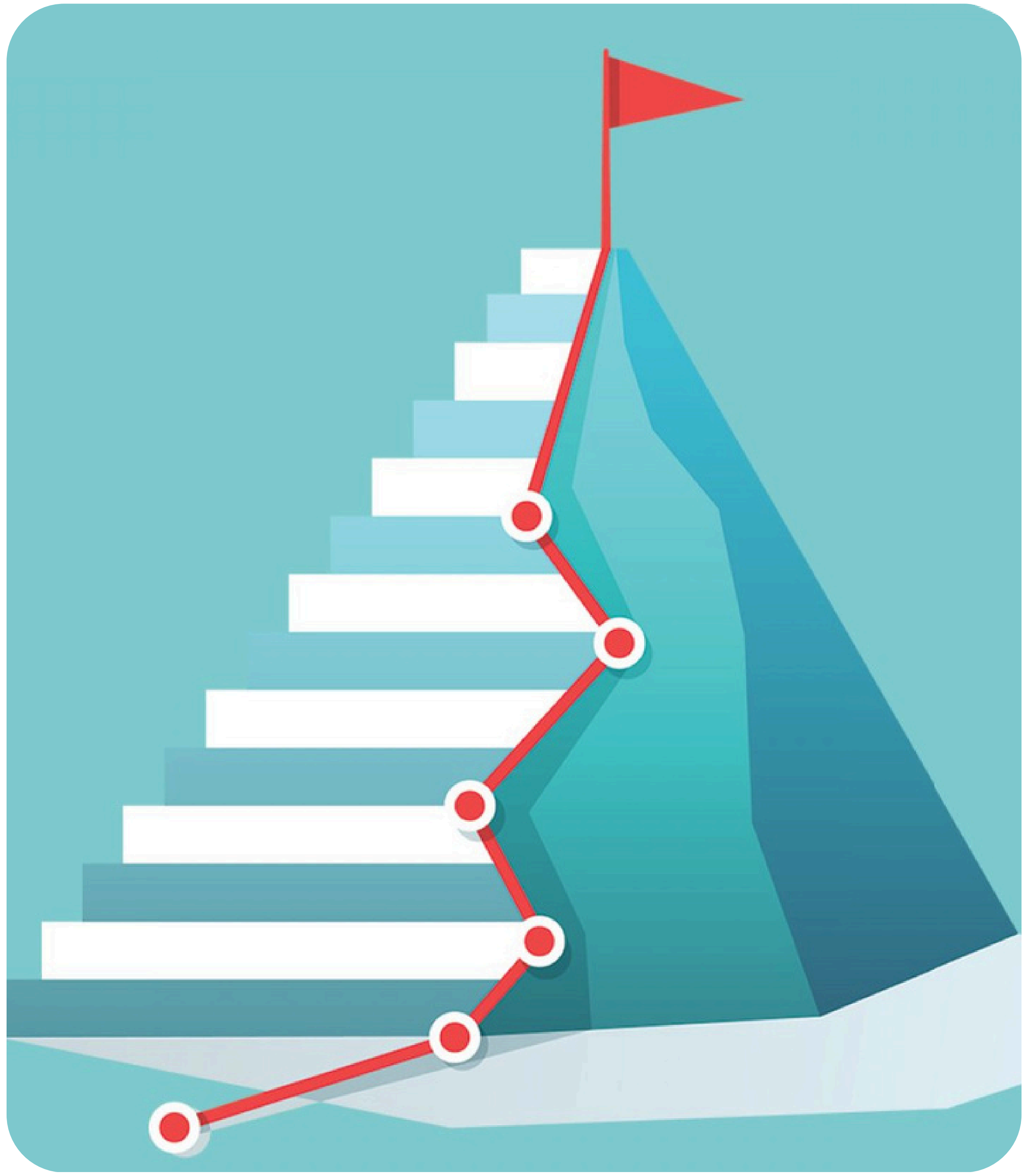
Déploiement professionnel

- Docker multi-stage Rust
- docker-compose (API + DB)
- Images légères
- Variables d’environnement
- Tunnel HTTPS possible

Compétences RNCP couvertes

- Modélisation Merise & SQL relationnel
- Développement d’API REST sécurisée
- Sécurité : hachage, JWT, cookies
- Tests, conteneurisation, déploiement

Conclusion



Travail fourni

- Système de connexion
- Système de capture dynamique
- Projet complet & opérationnel

Apprentissage

- Découverte d’Axum
- Approche professionnel
- Sécurité approfondie

Futur

- Ajout d’une CI/CD
- Ajout des features